

Mrunalini Ramnath

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EDUCATION

Carnegie Mellon University

Master of Science in Artificial Intelligence & Innovation

Pittsburgh, PA

Expected May 2027

- Coursework: Machine Learning (PhD), Deep Learning, AI Engineering, Adv. Natural Language Processing, Multimodal Machine Learning, Data Structures and Algorithms

Birla Institute of Technology & Science, Pilani

B.E. in Electronics & Communication Engineering; M.Sc. in Physics

Hyderabad, India

Jun 2022

- Coursework: Artificial Intelligence, Object-Oriented Programming, Operating Systems, Digital Image Processing

PROFESSIONAL EXPERIENCE

American Express

Senior Analyst - Credit & Fraud Risk

Bengaluru, India

Feb 2024 - Jun 2025

- Led the migration of large-scale ML feature pipelines from IBM mainframe to a proprietary cloud platform supporting online credit risk decisioning systems through rolling canary deployments while maintaining 99.9% production uptime
- Built and maintained PySpark-based data ingestion and validation pipelines for model features, enhancing reliability of downstream training and inference workflows
- Applied feature selection, dimensionality reduction, and statistical clustering to compress feature space by 43%, improving pipeline efficiency and throughput
- Delivered feature updates using data lineage and correlation analysis to elevate data quality for production ML models

Analyst - Credit & Fraud Risk

Jul 2022 - Jan 2024

- Optimized ML feature pipelines using tree-based selection, correlation filtering, and mutual information, cutting down feature usage by 32%
- Engineered automated PySpark and HiveQL validation frameworks for production ML datasets, saving 2500+ QA hours per quarter
- Facilitated large-scale model feature generation pipelines delivering \$15M annual business impact

Analyst Intern - Credit & Fraud Risk

Jan 2022 - Jun 2022

- Created high-volume feature datasets using PySpark and Hive to aid internal risk systems and ML workflows.
- Automated data tracking systems' triggered alert data retrieval and reporting, saving over 1000 hours per quarter
- Decreased false alert rate by 13% measured weekly by upgrading feature logic and alert filtering.

Computer Vision Lab, University of Ljubljana

Machine Learning Research Intern

Ljubljana, Slovenia - Remote

Jun 2021 - Dec 2021

- Integrated a CNN-based segmentation module into a biometric eye recognition pipeline and removed redundant stages.
- Benchmarked model variants, including U-Net, MobileNet, and DenseNet, for vascular-feature-based identification.
- Slashed inference latency to 27% by consolidating into a single compact model, with no material accuracy regression on internal benchmarks.

Adobe

Software Development Intern - Operational Security

Bengaluru, India - Remote

May 2021 - Jul 2021

- Assessed vulnerabilities across 10+ Kubernetes clusters, documenting 15+ critical security gaps.
- Benchmarked Kubernetes security tools (kube-bench, kube-hunter, AquaSecurity, kube-score, kubesecc), advancing coverage by 26%.
- Proposed remediation strategies reducing monitoring overhead by 15% with seamless integration.

PROJECTS

Cross-Modal Bottleneck Transformers for Efficient Multimodal Fusion | CMU

Oct 2025 - Dec 2025

- Designed and implemented multiple Multimodal Bottleneck Transformer architectures on CMU MOSI and CMU MOSEI, including early-, mid-, late-fusion, and bottleneck-based baselines
- Proposed a hierarchical bottleneck fusion mechanism that improved Acc2 by 4.3% and F1 by 2.8% over the strongest Multimodal Bottleneck Transformer baseline
- Performed systematic ablations over bottleneck placement and token budgets using reproducible training pipelines, quantifying accuracy-compute tradeoffs via FLOPs analysis
- Ensured experimental robustness by reproducing results across multiple random seeds and dataset splits; built config-driven training and logging pipelines to support extensible ablations

TECHNICAL SKILLS

Programming Languages & Tools: Python, C, C++, SQL, HiveQL, MATLAB, TypeScript, HTML/CSS, Git

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, PySpark, Apache Spark

Frameworks & Libraries: HuggingFace Transformers, LangChain, vLLM, OpenCV, NLTK, Django, Flask

Cloud & Platforms: AWS, Azure, Kubernetes, Docker

Machine Learning Systems: Model training pipelines, experiment tracking, reproducibility, scalable data processing